GENERAL BIOLOGY 2 (BIO 102)

DIFFERENCES AND SIMILARITIES BETWEEN PLANTS AND ANIMALS

There are many differences between plants and animals. When we say the plant, you think of green things which can prepare their own food. Similarly, when you hear animal, you think of organisms having a well-developed body with a proper organ system. Thus, they are easy to differentiate. However, there is a lot of difference between plants and animals which can be observed at a physical as well as the cellular level. Plants and animals fall under the category multicellular, eukaryotes. They share the same ecosystems, surroundings and their dependability on each other.

Definition of Plants

Plants fall under the multicellular, photosynthetic eukaryotes category. Further, they belong to the kingdom Plantae. There are different kinds of plants found in different areas and climates. Moreover, they are the primary producers on the Earth's ecosystem.

We call plants autotrophs because they are capable of producing their own food through photosynthesis. Generally, plants consist of a proper root system and the shoot system. The root system is one found below the ground.

Whereas, the shoot system contains parts above the ground like flowers, fruits, leaves, buds and more. There are different types of plants. They range from angiosperms, gymnosperms, ferns, conifers and mosses to liverworts, hornworts, green algae and more.

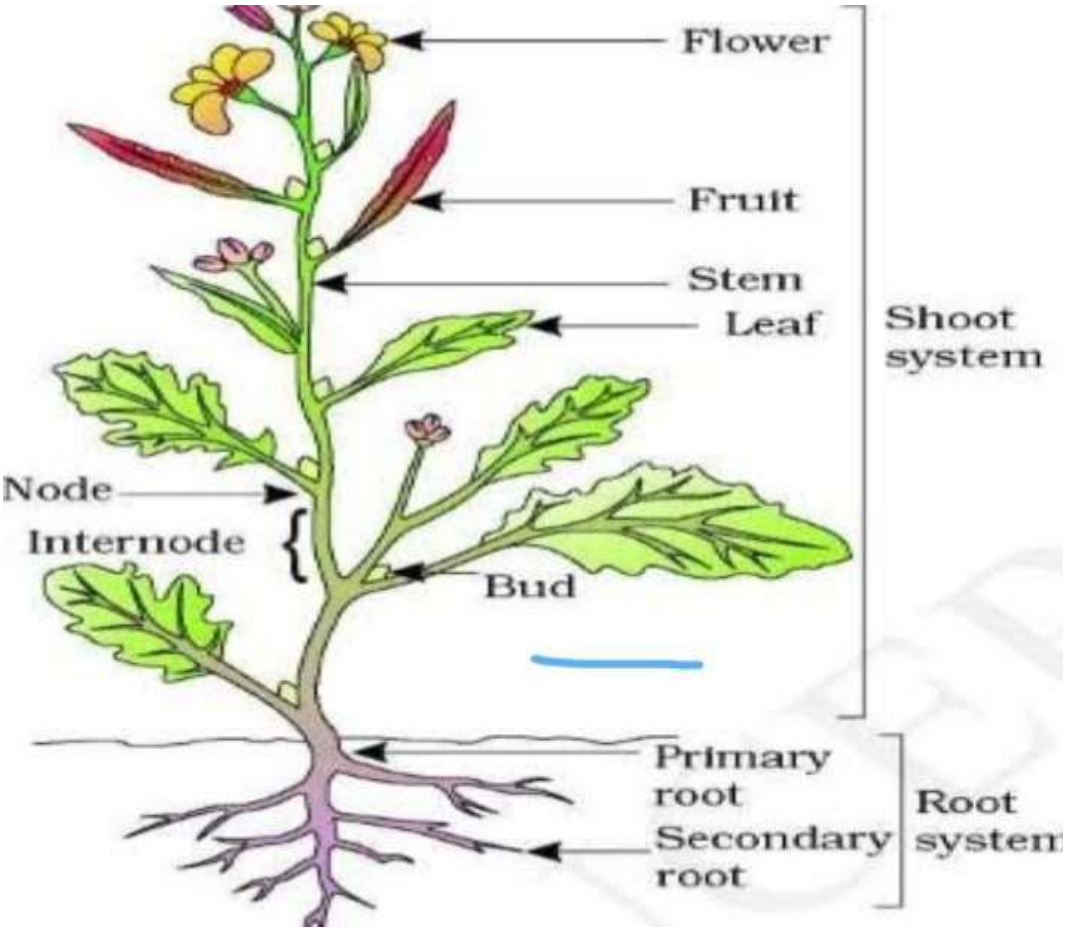


Fig 1. A diagrammatic representation of a typical plant

Definition of Animals

Belonging to the kingdom of Animalia, animals also vary from one place to another. Further, they fall under the category of multicellular, eukaryotes. Moreover, they are found in air, water and land.

Animals also have sub-divisions. For instance, they may be vertebrates or invertebrates, oviparous or viviparous, and even cold-blooded or warm-blooded. Generally, the most common groups of animals are classified as mammals, amphibians, reptiles, fish, birds and insects.

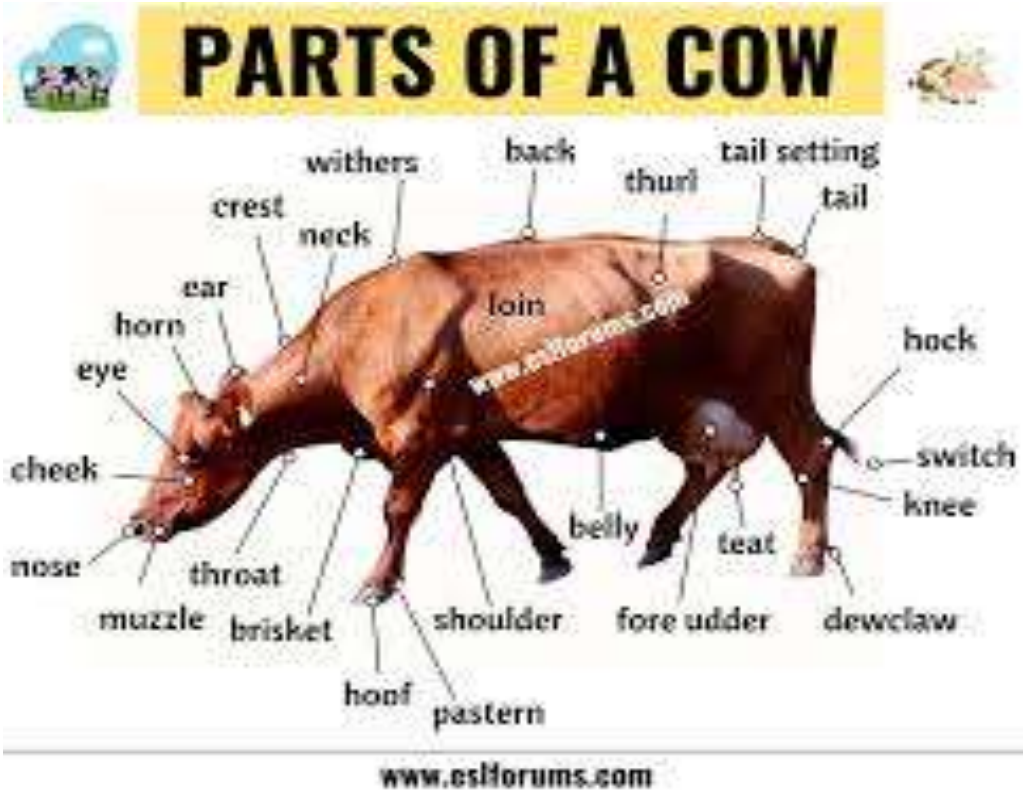


Fig 2. A diagrammatic representation of a typical Animal

Similarities Between Plants and Animals

Plants and animals are living things. They feed, respire, excrete, grow, move, reproduce and are sensitive to their environment. Animals and plants need food for energy but they feed in different ways. Animals eat plants and other animals, but plants make their own food.

Important Differences Between Plants and Animals

S/N	COMPARISON	PLANTS	ANIMALS
1	**Movement**	Cannot move as they are rooted in the ground. Exceptions- Volvox and Chlamydomonas.	Can move freely from one place to the other. Exception- Sponges and Corals.
2	**Digestive System**	Absent	Present
3	**Food Storage**	Store food in the form of starch	Store food in the form of glycogen
4	**Structure of Cell**	Contains cell wall, chloroplast, plasmodesmata, plastids and other organelles	Do not have cell walls, have other organelles like tight junction and cilia.
5	**Respiration**	Take in Carbon Dioxide and release Oxygen	Take in oxygen and release carbon dioxide
6	**Respiration through**	Occurs through stomata	Occurs through lungs, gills, skin and more.
7	**Growth**	The meristematic system in the tip of roots and stems supports growth	Organs and organ system support the growth
8	**Reproduction Method**	Asexual reproduction by budding, vegetative methods, wind, spores, insects.	Animals reproduce sexually whereas some lower animals like algae have asexual reproduction
9	**Response**	Show response through touch and light	The proper nervous system allows responding quickly

10	**Sensitive**	Less sensitive	Highly sensitive

Interdependency of Plants and Animals

1. Animals breathe oxygen and exhale carbon dioxide. Plants take in carbon dioxide and release oxygen back into the air.

2. Animals need plants for food and shelter.

3. Plants need animals for seed dispersal and pollination.

4. Some animals eat plants. Some animals eat other animals. When animals die and decompose, the plants use the nutrients that were deposited back into the soil from the decomposing organism.

5. Forests and wetlands filter water so it is clean, and animals need clean water to drink.

Dr. K.C Irikannu